Are the most preferred IISERs the ones with the fewest women?

The rank and preference pattern, tested on NIRF data for the seven IISERs, with IISc Bangalore as a reference point

Summary.
Question. Admission to the IISERs is by rank in a common national entrance test (the IAT) with no reservation for women. If the institutes students prefer are filled by the highest ranks, and boys dominate the top of the rank list, the most preferred institutes should have the fewest women.
Finding. They do. According to the NIRF ranking, the share of women among UG five-year students rises as an institute sits lower in the order: Spearman 0.82 across the seven IISERs (exact permutation p = 0.03). The order the interviews call student preference gives 0.86 (p = 0.02). Pune and Kolkata, first and second, have 27.6% and 27.2% women; Tirupati and Thiruvananthapuram, fifth and sixth, have about 52%. There is no such pattern for faculty (−0.32). IISc Bangalore, which has 16.9% women among its UG students and is placed first in the order, as the most preferred institute, gives 0.88 (p = 0.007).
How firm. ***This is simply a pattern, not proof***. Seven points, wide uncertainty, and rank is tied to age, size and region. The mechanism, who reaches high ranks and why, cannot be tested with public data.

1. Where the question came from

Dr. Anuradha Bhat showed us that women's share at the IISERs runs almost opposite to institute ranking and student preference: Pune and Kolkata are the most preferred and have the fewest women; Thiruvananthapuram, Tirupati and Berhampur are the least preferred and have the most. Because the entry test is national with no gender reservation, a gap at admission has to come from who gets high ranks. Her interview changed the direction of the study. Until then we had treated Kolkata as an isolated case and looked at location and home state; this suggested the gap begins well before IISER. When we showed the pattern to Dr. Rumi De she explained the likely cause: admission is rank-wise, the top of the list is dominated by boys, and **highly competitive exams need coaching and family investment, which families are often more willing to make for sons. She stressed that this is not about ability.**

2. Data and method

- Women as a share of UG five-year students, latest year for each institute, from the NIRF "Total Actual Student Strength" tables (Report 3): Pune 27.6%, Kolkata 27.2%, Bhopal 39.3%, Mohali 39.9% (2023-24), Tirupati 51.8%, Thiruvananthapuram 51.6%, Berhampur 43.7%. Women among regular faculty are from Report 4.

- Order of institutes: NIRF ranking order Pune, Kolkata, Bhopal, Mohali, Tirupati, Thiruvananthapuram, Berhampur. The interviews with Dr. Bhat quote a slightly different preference order: Pune, Kolkata, Bhopal, Mohali, Thiruvananthapuram, Tirupati, Berhampur. We ran both.
- IISc Bangalore: women are 16.9% of UG students (2024-25) and 10.0% of faculty (2026), from Reports 3 and 4. IISc is not an IISER, it accepts admissions via channels other than IAT as well, and its undergraduate programme is four years, so it is shown as a reference point (hollow marker, marked *). We place it first in the order because it ranks above all the IISERs in NIRF rankings. Moreover, it is generally the most coveted institute among students and one of the most highly regarded research institutes in India, so it is first in preference.
- Measure: Spearman rank correlation between position in the order and women's share. With only seven points the usual p-value is unreliable, so we also give the exact permutation p (all 5,040 orderings).

3. Result



Figure 1. Women among UG students (left) and regular faculty (right) against position in the ranking order. Orange: IISER Kolkata. Hollow: IISc Bangalore (* four-year programme). Source: NIRF; Reports 3 and 4.

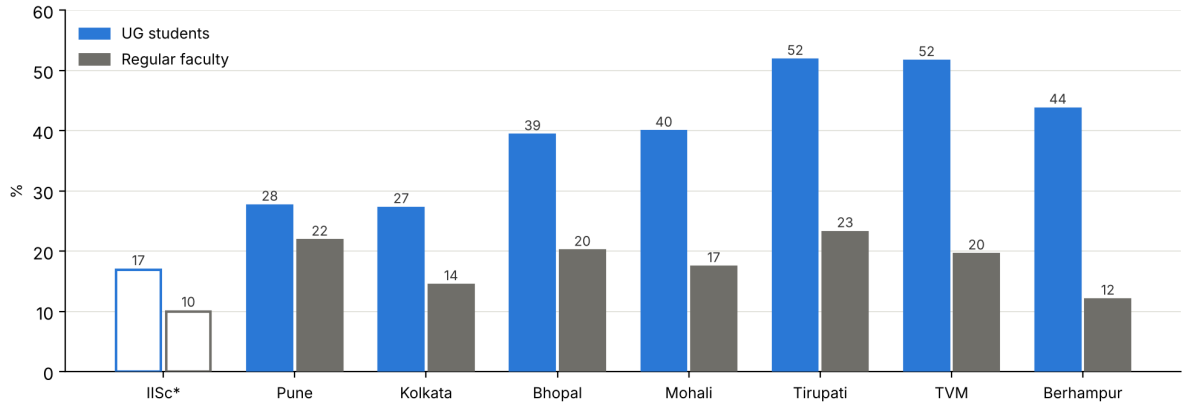


Figure 2. The same shares by institute, in ranking order. Students and faculty do not move together.

Order used	Women among UG students	Exact p	Kendall τ	Without any one institute	Women among faculty
NIRF ranking order	ρ = 0.82	0.03	0.62 (p = 0.07)	0.71 to 0.89	ρ = −0.32 (p = 0.48)
NIRF order with IISc first (8 institutes)	ρ = 0.88	0.007	0.71 (p = 0.01)	0.82 to 0.93	ρ = 0.12 (p = 0.78)
Preference order from the interviews (7 IISERs)	ρ = 0.86	0.02	0.71 (p = 0.03)	0.77 to 0.94	ρ = −0.21 (p = 0.64)

The two top-ranked institutes have the lowest shares (27.6% and 27.2%) and the two at the fifth and sixth places the highest (51.8% and 51.6%). The pattern is not perfect: Berhampur, last, has 43.7% (probably because it is a relatively new institute), lower than Tirupati and Thiruvananthapuram, and Mohali (39.9%) is level with Bhopal (39.3%). Dropping any single institute leaves the correlation between 0.71 and 0.94, so no one institute drives it. A rough 95% interval for the first correlation is wide (about 0.1 to 1.0). The faculty shares show nothing consistent: −0.32 for the seven IISERs, 0.12 once IISc is added, neither distinguishable from zero. That fits the view that **the gap at admission is a different matter from the gap in hiring.**

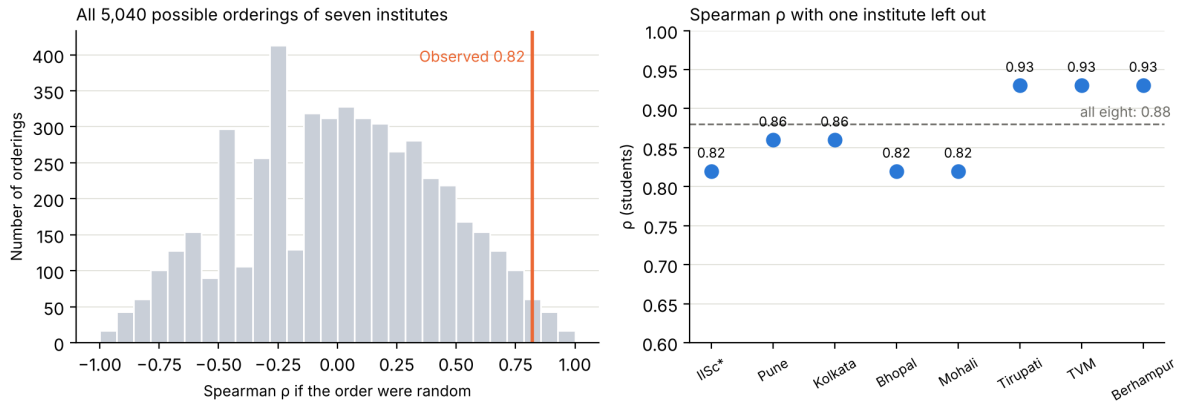


Figure 3. Left: where the observed 0.82 falls among all possible orderings of the seven IISERs. Right: the eight-institute correlation with each institute left out in turn.

What IISc does to the result

IISc has the lowest share of women among UG students (16.9%) and among faculty (10.0%) of all eight. Placed first, as the most preferred institute, it fits the pattern and lifts the correlation from 0.82 to 0.88. It is a reference point, not an IISER: its programme is four years and its entry route differs, so we treat it as consistent with the pattern, not as extra evidence for it.

4. What this can and cannot say

- **It is a pattern at institute level.** Seven (or eight) institutes are a small sample, and the students counted are five cohorts admitted over several years, while the ranking is a single year's.
- **Rank is tied to other things.** Ranking goes with an institute's age, size and region. The two southern institutes sit low in preference and high in women, and Dr. De suggested that families may find the

south safer and prefer an institute they can reach; Dr. Ta suggests parents may simply prefer some IISERs for their daughters. Safety and location could produce the same picture without ranks playing the main part. This data cannot separate the two.

- **The mechanism is not tested.** The explanation that fewer women reach the top ranks, and why (coaching, family investment, expectations), needs applicants, ranks and the institutes chosen, by gender. These are not public. Dr. Ta suggested that an RTI request would be needed.
- **It says nothing about ability.** The interviews are clear on this, and the grade data point the same way: women do as well as men once admitted (Reports 10 and 11).
- **The same pattern for the IITs** is mentioned in the interviews. We did not reproduce it for this report.

5. What would test it

- IAT applicants, qualifiers and allotted candidates by gender and rank band, and the institutes they chose in counselling.
- Students' state of origin by institute, which would also test the nearby-states idea raised by Dr. Bhadra and Dr. De.

Sources

National Institutional Ranking Framework (NIRF), Ministry of Education, Government of India. "Overall" category submissions, 2019 to 2026. nirfindia.org.
Team reports: Report 3 (women among IISER students), Report 4 (women among IISER faculty), Reports 10 and 11 (grades by gender). Interviews: Dr. Anuradha Bhat; Dr. Rumi De; Dr. Malancha Ta.

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